

Case Study 14

How does the final grade vary depending on preschool attendance, taking age into account?

1. Framework

This assignment aims to practically apply the contents of the course unit: **Data Analysis using the Python programming language**, object types and their handling, mathematical calculations, database importing, cleaning and organization, exploratory data analysis, and data visualization.

The assignment will be carried out in groups or individually, using Python, and must take the form of a commented notebook, accompanied by a brief oral presentation.

2. Dataset

All groups will work with the same database, corresponding to the student dataset.

Either the student-mat.csv file or the student-por.csv file may be used.

The database includes demographic, family, school, behavioral, and academic performance variables, including the grades G1, G2, and G3.

The database used in this assignment corresponds to the *Student Performance dataset*.¹

3. Final Deliverables

- 1 commented Python notebook (.ipynb).
- 5 minute oral presentation.

4. Mandatory Assignment Structure

The notebook should be organized into 4 main tasks :

Task 1 - Reading, Selecting, and Organizing the database

Task 2 - Descriptive Statistics

Task 3 - Graphical visualization

Task 4 - Interpretation and Conclusion

5. Oral Presentation

Each group will give a 5-minute oral presentation, in which they should present the assigned topic, the chosen variables, the main statistical results, the graphs produced, and the main conclusion.

The presentation should be brief, objective, clear, and focused on the main results.

¹ Cortez, P. (2008). *Student Performance* [Dataset]. UCI Machine Learning Repository.